RESEARCH PROPOSAL

Research title ***(English)***	Beyond Forecast Accuracy: Comparing Probabilistic Demand Forecasts by the Inventory They Require across Retail Settings and Periods
Sub-committee	Information Technology
Group name	
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Abstract

Public retail forecasting benchmarks rank methods by forecast error, although forecasts are used to decide how much stock to hold. This research asks which probabilistic demand forecasting method needs the least inventory to reach a target service level in a realistic replenishment policy, and whether the answer holds across retail settings and time periods. The problem is that studies linking forecasts to inventory decisions usually rely on one dataset, one test period, selected series and assumed cost parameters, so their conclusions may not transfer to other retailers or periods. We embed seven probabilistic forecasting methods (three statistical, four gradient-boosting) and a neural baseline in one order-up-to policy with lost sales and optional liquidation, and evaluate them on all eligible series of two public datasets, M5 (brick-and-mortar) and VN1 (multi-vendor e-commerce), across three 26-week test windows. Methods are compared by the inventory they require at the same realised fill rate, with bootstrap confidence intervals, complemented by regret, a cost view over lost-sales penalties, and sensitivity analyses of lead time and hyper-parameters. Preliminary results show that LightGBM quantile regression is the most efficient single method on M5 in every window, whereas on VN1 the best method changes with the period; a post hoc equal-weight quantile combination of LightGBM and a negative-binomial TSB model is the most robust choice in the tested windows and will be validated on new periods.

Key words: probabilistic forecasting; inventory management; intermittent demand; quantile regression; forecast combination; M5; VN1

1. Introduction

1.1. Literature review

Retail demand forecasting has been driven by public competitions. In the M5 accuracy track, LightGBM [1] was used by all of the top 50 teams and the winning solution combined LightGBM models trained with a Tweedie loss [2]; the winner of the uncertainty track trained a separate LightGBM model for each quantile [3]. These competitions score forecasts by error measures such as the weighted scaled pinball loss [4], and their organisers state that they did not target a specific decision problem [3]. The more recent VN1 e-commerce dataset [5] has likewise been used to study accuracy and computational cost [6]. Forecast accuracy, however, is not the quantity a retailer pays for: a structured review observes that most forecasting studies ignore the step from forecast to replenishment decision [7].

A growing body of work evaluates forecasts by their inventory consequences. Kourentzes et al. tune exponential smoothing with an inventory-based cost function [8]; van der Haar et al. learn order quantities end to end [9]; Petropoulos et al. simulate an order-up-to policy over rolling origins on M3 data and find that forecast combinations perform best [10]. On M5, Spiliotis et al. compare 21 probabilistic methods in a 28-day daily simulation and find that simple empirical quantile methods outperform machine learning in inventory terms [11]; Wang et al. study probabilistic combinations with a single-period newsvendor rule [12]; Zabraoui et al. simulate reinforcement-learning policies on selected items [13]; and Mohammed et al. evaluate safety stock on 8,000 high-volume series [14]. The closest study, by Theodorou et al. [15], simulates one year of an order-up-to policy with lost sales on 7,783 daily product–state series of M5 for 13 methods whose quantiles are point forecasts plus empirical error quantiles. Trade-off curves broadly follow accuracy, but total costs that weight lost sales heavily favour less accurate methods, and the authors call for replications on online retail data.

Two further strands are relevant. For intermittent demand, the TSB method updates a demand probability every period [16], demand is classified by the average inter-demand interval and the squared coefficient of variation (ADI–CV^2) [17], the negative binomial is a common distribution for stock control [18], and no global architecture is established for such series [19]. For model choice, forecast combinations hedge against model uncertainty, and the "forecast combination puzzle" states that equal weights often beat estimated weights [20]; feature-based meta-learning such as FFORMA learns combination weights from series features [21]. Neural and foundation models such as N-HiTS [22] and Chronos-2 [23] are increasingly used as global forecasters.

Quality and gaps. The reviewed sources are peer-reviewed journals and proceedings (IJF, EJOR, IJPE, AAAI) or preprints from established groups; the preprints are marked as such. Four gaps and inconsistencies emerge. (1) Studies on M5 often keep only selected series, such as high-volume items [14], food items [13] or series with at least 2.5 years of history [15]. (2) Most use one dataset and one test period [12], [15], [11], so it is unknown whether conclusions transfer across settings and periods; Theodorou et al. themselves caution that one simulated year gives no guarantee for the next [15]. (3) Cost-based evaluations depend on assumed parameters, and cost-based and cost-free conclusions can disagree [15]. (4) Results conflict: simple methods win in daily simulations [15], [11], whereas LightGBM dominates the M5 competitions [2], [3]. Our literature search has not yet identified studies by Vietnamese researchers that evaluate probabilistic retail forecasts by inventory decisions; the search will be extended in local journals and conferences during the project.

Applications and implications. Decision-based evaluation matters for retailers who must choose a forecasting method without cost data or a data-science team. The implication for research is that benchmarks should report inventory at equal service level over several periods; for practice, that a method choice should be robust when the best method cannot be known in advance.

1.2. The necessity of the research

Problem. A retailer needs to know which forecasting method keeps the least stock for a required service level, but public benchmarks answer a different question (which method has the lowest error), and decision-based studies are limited to one setting and one period. The research tests three hypotheses: H1 – the ranking of methods by inventory at equal realised fill rate differs from the ranking by forecast error; H2 – the most inventory-efficient method depends on the retail setting and on the test period; H3 – a combination of complementary methods is more robust (smaller worst-case regret) than any single method.

Originality and significance. The research combines, for the first time to our knowledge, (i) two public retail settings, brick-and-mortar and e-commerce, (ii) all eligible series including intermittent ones, (iii) three test periods, (iv) comparison at equal realised fill rate rather than at equal nominal service target, and (v) a test of whether demand classes, unsupervised segmentation or a growth predictor can choose the method in advance. The scientific value lies in evidence on when forecast accuracy translates into inventory efficiency; the practical value lies in a simple, reproducible recommendation for small retailers.

1.3. Feasibility of research

+ **Research question:** Which probabilistic forecasting method, or combination, needs the least inventory to reach a target fill rate in an order-up-to policy with lost sales, and does the answer hold across retail settings and test periods? Sub-questions: RQ1 – how do statistical, gradient-boosting and neural methods compare at equal realised fill rate on M5 and VN1; RQ2 – why does the best method change between periods, and can demand classes, unsupervised segmentation or a growth predictor anticipate it; RQ3 – is a quantile combination robust, and how does the answer depend on the cost of a lost sale; RQ4 – does liquidation reduce inventory without losing service, and at which salvage price does it break even under assumed costs.

+ **Research design:** An empirical benchmark with rolling-origin forecasting and multi-period inventory simulation. Each method forecasts quantiles {0.5, 0.8, 0.9, 0.95, 0.99} of cumulative demand over the protection interval directly; the policy orders up to the τ-quantile; the inventory needed at each fill target is interpolated along the inventory–fill-rate curve; statistical significance uses bootstrap intervals over series and the Friedman–Nemenyi test [24].

+ **Sample size and population:** The population is retail SKU–location demand series. Sample: M5 contains 30,490 product–store series over 277 weeks; VN1 contains 15,053 client–warehouse–product series over 196 weeks. Inclusion criterion: a series starts at least 13 weeks before the test window and has at least one sale before it, giving 30,381 / 29,917 / 28,824 eligible M5 series and 13,844 / 11,442 / 9,383 eligible VN1 series in the three windows. No further sampling is applied.

+ **Data sources:** Existing public datasets only: M5 (Kaggle competition data) and VN1 (DataSource.ai). No surveys or interviews are needed. The datasets are used under their terms and are not redistributed.

+ **Budget and funding:** No direct cost: data are public and all experiments run on the authors' laptops (Intel Core i5-12450H, 16 GB RAM, CPU only). A possible conference registration fee for the accompanying paper is outside this proposal.

+ **Timeframe:** October to December 2026, aligned with the competition milestones (Section 5). The core benchmark and robustness analyses are already running and reproducible from a public repository (https://github.com/anhduyBA/forecast-to-decision-benchmark).

+ **Potential challenges and risks:** (1) Sales are used as demand, so historical stock-outs are invisible – mitigated by stating the limitation and by comparing methods under identical data. (2) The combination was specified after the main analysis – mitigated by validating it on new periods before claiming it. (3) Bootstrap intervals resample series, not periods – mitigated by three separate test windows. (4) Long computation on M5 – mitigated by caching forecasts and splitting runs across two machines with data fingerprints. (5) Dataset differences may confound comparisons – mitigated by re-weighting, price ablation and shared-feature checks.

2. Research objectives

O1 – Build a reproducible benchmark that embeds probabilistic forecasting methods in one order-up-to policy with optional liquidation, on all eligible series of M5 and VN1, repeated on three 26-week test windows.

O2 – Define and apply a cost-free evaluation protocol: inventory at equal realised fill rate with bootstrap confidence intervals, frontier rank and regret, complemented by a cost view over lost-sales penalties.

O3 – Explain differences between settings and periods, and test demand classes, an unsupervised segmentation, a growth-prediction rule and quantile combinations as ways to choose a method.

O4 – Check robustness to lead time, hyper-parameters, price features, feature parity and a neural baseline.

O5 – Validate the recommended combination on periods not used to select it, and publish code and results.

3. Research scope

Included: weekly demand of SKU–location series in two public retail datasets; seven probabilistic forecasting methods (empirical quantiles, exponential smoothing, TSB with a negative-binomial distribution, LightGBM with Tweedie loss plus safety stock, LightGBM with conformal quantiles, histogram gradient boosting and LightGBM quantile regression), a univariate N-HiTS baseline and quantile combinations; an order-up-to policy with weekly review, lead times of one, two and four weeks and lost sales; four liquidation rules evaluated over 26 weeks.

Excluded: price-response (promotion) modelling, multi-echelon inventory, pack-size and minimum-order constraints, hyper-parameter tuning beyond the sensitivity analysis, and deployment at a real company. Conclusions are limited to the tested datasets, windows and policy settings.

4. Approach and Method

RQ1 – Benchmark. Each method forecasts quantiles of the h-week total $D_h(o) = Y[o] + \dots + Y[o+h-1]$ directly ($h = 3$ for replenishment, $h = 13$ for liquidation), using a rolling origin over the last 26 weeks in two 13-week blocks with refits. The global models (LightGBM [1], quantile regression [25], conformal quantiles [26], scikit-learn [27]) share lag, rolling, calendar and price features; the statistical models follow [28], [16]. Accuracy: volume-weighted quantile loss, coverage and per-series ranks with Friedman–Nemenyi [24]. Inventory: the policy orders $\max(S - IP, 0)$ with S the τ-quantile; τ runs over the five quantile levels, and the inventory needed at fill targets 0.90–0.98 is interpolated along each method's curve, with 200 bootstrap resamples of series.

RQ2 – Period dependence. Series are split ex post into stable and growing demand to explain rankings; prospective selection is then tested with ADI–CV^2 classes [17], a Gaussian-mixture segmentation chosen by pre-defined stability criteria, and a classifier that predicts growth from pre-test features, each learned on an earlier window and applied to the next. Comparability between datasets is checked by inverse-propensity re-weighting, price ablation and refits with shared features only.

RQ3 – Robust choice. Equal-weight quantile averaging [20] of complementary methods is compared with single methods and with learned weights; regret is the excess inventory over the best single method per run. A cost view maps each curve to cost $C(r) = \text{inventory} + r \cdot (1 - \text{fill rate})$ for lost-sales penalties r from 10 to 1,000 weeks of holding. Robustness: lead times of one, two and four weeks in every window, four LightGBM settings, and a neural baseline (N-HiTS [22]).

RQ4 – Liquidation. Quantile-based, fixed and dead-stock rules are simulated with the same forecasts; the break-even salvage ratio is computed over holding rates of 10–40% per year and margins of 30–100%.

Validation of the combination (O5). The combination will be re-evaluated on test windows that were not used to choose it, so that the robustness claim does not rest only on the windows in which it was specified.

5. Research plan

The research plan, with expected outputs, schedule and person in charge, is given below.

No.	Date	Task	Output	Person in charge
1	15/09/2026 – 05/10/2026	Literature review; data acquisition and harmonisation of M5 and VN1	Literature matrix; weekly panels; data checks	Bùi Anh Duy, Khổng Trần Gia Huy
2	01/10/2026 – 12/10/2026	Benchmark of seven methods on three test windows	Forecasts, simulation results, accuracy and inventory tables	Bùi Anh Duy
3	10/10/2026 –	Robustness: hyper-parameters,	Sensitivity tables;	Bùi Anh Duy,

	16/10/2026	lead time, N-HiTS baseline, cost view, dataset checks	neural baseline results	Khổng Trần Gia Huy
4	13/10/2026 – 20/10/2026	Research proposal and conference draft	Proposal (this document); draft paper	Bùi Anh Duy, Khổng Trần Gia Huy
5	21/10/2026 – 30/10/2026	Revise and submit the full paper to ICIIT 2027 (approved by the mentors)	Submitted conference paper; public code repository	Bùi Anh Duy, Khổng Trần Gia Huy
6	31/10/2026 – 01/11/2026	Qualifying round presentation	Presentation slides	Bùi Anh Duy, Khổng Trần Gia Huy
7	02/11/2026 – 27/11/2026	Validate the combination on unseen periods; extend the literature search; address review comments	Validation results; revised manuscript	Bùi Anh Duy, Khổng Trần Gia Huy
8	28/11/2026 – 07/12/2026	Full paper for the competition in the IEEE template	Full research paper	Bùi Anh Duy, Khổng Trần Gia Huy
9	08/12/2026 – 20/12/2026	Final presentation preparation and final round	Slides; final presentation	Bùi Anh Duy, Khổng Trần Gia Huy

The mentors, Nguyễn Thị Cẩm Hương and Trương Long, review the design, results and manuscripts at each milestone.

6. Expected results

Theoretical implications. Evidence on when forecast accuracy translates into inventory efficiency, and on how the answer depends on the retail setting, the test period and the cost of lost sales. Preliminary results (Appendix A) indicate that LightGBM quantile regression is the most efficient single method on brick-and-mortar data in every window and lead time tested, whereas on e-commerce data the best method changes with demand growth.

Methodological implications. A cost-free protocol, inventory at equal realised fill rate over several periods, that avoids rewarding methods whose quantiles under-cover, together with regret and cost-view summaries; and a negative result: demand classes and unsupervised segmentation do not choose the method reliably.

Practical implications. A simple, reproducible recommendation for retailers without cost data: an equal-weight quantile average of a global quantile model and a statistical intermittent-demand model,

used unless lost sales are very expensive, subject to the planned validation. The benchmark code and results are public.

Plan to apply the research in practice. The recommended models will be served in RetailIQ, the team's capstone warehouse-management system, which computes replenishment suggestions from probabilistic forecasts with an order-up-to policy and requires human approval of every order.

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Appendices

Appendix A. Preliminary results (from the public repository)

Analysis	Preliminary finding
Single method with the lowest inventory at equal fill rate, M5	LGB-Q in all three windows; 3.3–10.6% less inventory than LGB-T, LGB-C, TSB-NB and ETS at fill rate 0.94
Same, VN1	Changes with the window (LGB-Q, TSB-NB, LGB-Q)
Lead times 1, 2 and 4 weeks (18 runs)	M5: LGB-Q ahead of LGB-T, LGB-C, TSB-NB and ETS in all nine window × lead-time runs
Quantile combination C2 (post hoc)	VN1: 3.2–10.9% less inventory than LGB-Q at fill rate 0.92 across windows and lead times; maximum regret 2.0% vs 12.6% for LGB-Q
Hyper-parameter settings	M5 conclusions hold under four settings; VN1 under three of four
Neural baseline (N-HiTS)	Well calibrated; not the most inventory-efficient method in any run
Liquidation	Quantile rule costs ≤ 0.02 percentage points of fill rate; no economic benefit within 26 weeks under assumed costs

These results come from three test windows; the combination was specified after the main analysis and is validated on unseen periods in the research plan.

COMMITMENT

ON RESPONSIBLE ARTIFICIAL INTELLIGENCE USAGE IN RESEARCH

We are authors of the research paper titled "Beyond Forecast Accuracy: Comparing Probabilistic Demand Forecasts by the Inventory They Require across Retail Settings and Periods", would like to ensure that all members comply with the rules relevant to Artificial Intelligence (AI) regulations and ethical guidelines in research practices, which are described in details as follows:

1. AI is only used as assistant in correcting grammar, structure and phrasing in academic writing;
2. AI usage should NOT be engaged with the creation of content or drafting the research paper, production of fake data or literature synthesis;
3. AI usage should be checked and confirmed by all authors. Each author shall bear full responsibility in the accuracy, transparency and originality of their sections, including citations and reference;
4. All authors should disclose the use of generative AI and AI-assisted technologies to other co-authors, helping them understand where and how AI was applied in research preparation.

Disclosure of AI use (rule 4): A generative AI assistant (Claude, Anthropic) was used in this project to write and debug analysis code, to run experiments on the public datasets, and to prepare drafts of this proposal and of the accompanying paper, including summaries of the cited literature prepared from full-text copies obtained by the authors. All numbers come from experiments that the authors can reproduce from the public repository. The authors have reviewed every section and take full responsibility for its accuracy. Because rule 2 restricts AI-generated content, the authors will confirm with the organising committee how this use should be handled before signing.

Hereby, we would like to confirm and take full responsibility for the accuracy and transparency of the provided details.

Author 1	**Author 2**	**Author 3**
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